

7 Methicillin-resistant *Staphylococcus aureus* (MRSA) is a disease causing worldwide concerns.

Methicillin is one of the antibiotics to which this pathogen has developed resistance.

Castaneroxy A is a molecule extracted from the European chestnut tree, endemic to Southern Europe and Turkey.

The effects of Castaneroxy A have been investigated.

The investigations have found that Castaneroxy A:

- does not affect the natural skin flora
- prevents MRSA from producing toxins
- weakens the MRSA bacteria
- reduces the size of skin infections caused by MRSA.

(a) Explain why the development of this drug is an example of an 'evolutionary race' between humans and this pathogen.

(2)

(b) Explain why it is important that Castaneroxy A does not affect skin flora.

(2)

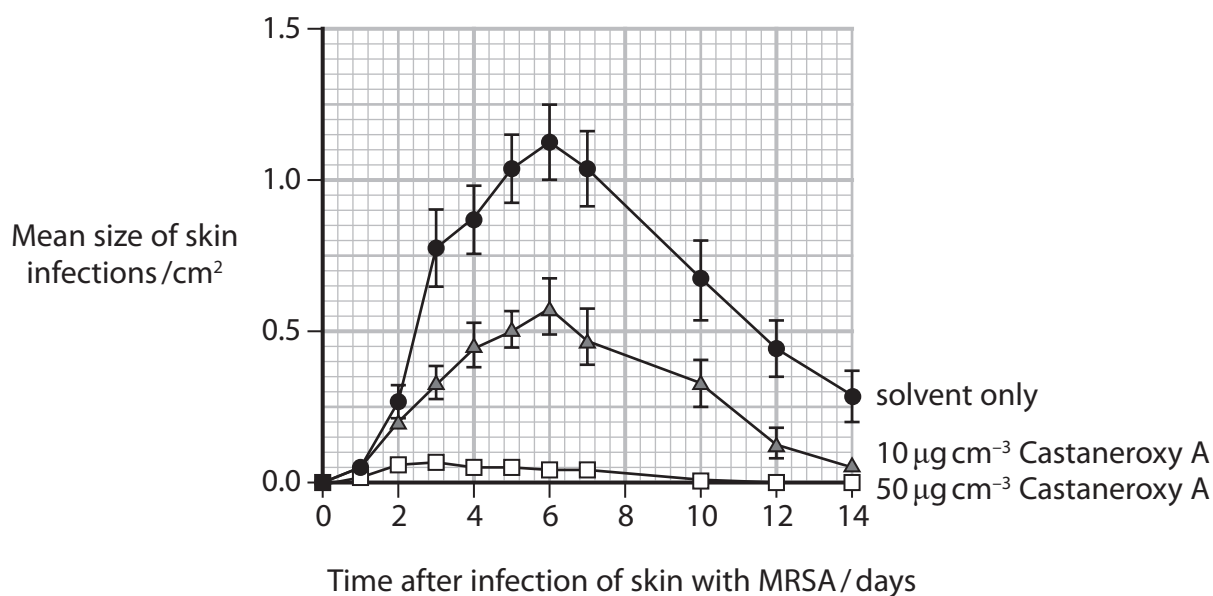
- (c) Suggest how weakening the MRSA bacteria could help the recovery of a patient infected with this bacteria.

(2)

- (d) In an investigation, patients who had developed skin infections were divided into three groups.

Two groups were each treated with a different concentration of Castaneroxy A. The other group was treated with only the solvent that was used in the treatment.

The graph shows the results of these treatments on the mean size of the skin infections.



(i) Explain why this investigation included treatment with the solvent only.

(2)

(ii) Determine the effect that Castaneroxy A has on the mean size of skin infections.

(3)

(iii) The leaves of the chestnut tree contain 0.0019 %, by mass, of Castaneroxy A.

Calculate the mass of leaves, in kilograms, that would be needed to produce 1 dm^3 of Castaneroxy A at a concentration of $50 \mu\text{g cm}^{-3}$.

Give your answer to the nearest kilogram.

(3)

Answer kg

(Total for Question 7 = 14 marks)